# Fuselage Section 13 — Forward — Door Surround Structure

Fwd Cargo Door Surround Fitting — Corner Crack Stop Drill and Reinforcing Angle

| | | | |
|---|---|---|---|
| **Object code** | DMC-SLN-A-53-21-13-00A-664A-A | **Issue** | 001-00 (new) |
| **Issue date** | 1996-06-20 | **Language** | en-GB |
| **Responsible partner** | AERALIS S.A.S. (U8025) | **Originator** | AERALIS S.A.S. — Repair Design, Neuhafen (U8025) |
| **Security** | 01 — Unclassified | **Quality assurance** | First verification (tabtop) |
| **Applicability** | SOLANO-30 and SOLANO-34, all aircraft | | |

## 1. Reason for the job

This repair arrests a crack at the lower corner of the forward cargo door surround fitting and restores enough static capability for continued operation until the fitting is replaced at the next opportunity.

## 2. Job set-up information

### A. Referenced information

| TYPE | DESIGNATION | IDENTIFICATION No. |
|---|---|---|
| GEN | Cracks up to 18 mm measured from the radius tangent point, originating at the corner radius, not crossing a fastener row and not branched. One crack per fitting. | — |
| GEN | The damage is in zone 111, 121. | — |
| GEN | A repair approval request has been raised with Aeralis. | — |

### B. Fixtures, tools, test and support equipment

*No support equipment required*

### C. Consumable materials

| QTY | DESIGNATION | IDENTIFICATION No. |
|---|---|---|
| 36 EA | WASHER, COUNTERSUNK, ALUMINIUM | AES963502K7 (CAGE U8025) |
| 36 EA | WASHER, PLAIN, CORROSION RESISTANT | AGS46790-6A (CAGE U8025) |

### D. Expendable parts

*No expendable parts required*

## 3. Procedure

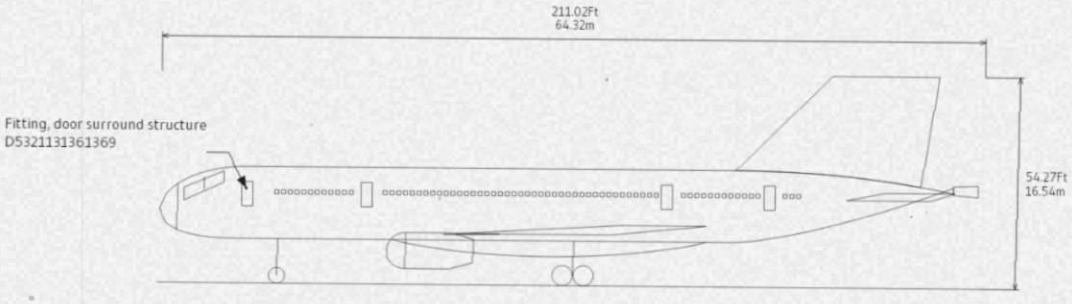
**1. Damage assessment**

Find the crack tip by dye penetrant inspection and mark it. Cracks up to 18 mm in length, originating at the corner radius and not crossing a fastener row, are inside this repair. A crack that has reached a fastener hole, that has branched, or that lies in the frame sub-assembly rather than in the fitting, is outside it and requires an engineering assessment.

## 3. Procedure

2. Measure the crack length from the radius tangent point, not from the visible surface break, because the subsurface extent is typically 3 to 5 mm longer. Do an eddy current inspection of the opposite corner of the same fitting before deciding on the repair; symmetrical cracking usually means the load path has changed and a single-corner repair will not hold.

3. See Fuselage Section 13 — Forward — Door Surround Structure — location, side view.




**Fuselage Section 13 — Forward — Door Surround Structure — location, side view — Figure 1**

### 4. Repair procedure

Stop drill the crack 4 mm beyond the marked tip with a 4.8 mm drill, and ream the hole to remove the drilled surface. Do a dye penetrant inspection of the stop drill hole to confirm the crack tip is contained. Cut the reinforcing angle from the fitting kit D5321131361369 material to the layout of repair drawing RD-53-21-13-003 and drill the fastener pattern using the angle as a template.

5. The inboard row is set out 12 mm from the sealant land at the frame foot, which is the datum for every dimension on sheet 2. Do not drill into the frame sub-assembly D5321134517524; the drawing gives the exclusion boundary. Apply the chromate-free primer to all new surfaces.

6. Install the angle wet with sealant and fasten it with washers AES963502K7 and plain washers AGS46790-6A at the pitch on the drawing. Fit silicone edging grommets where the angle crosses a systems penetration. Restore the continuous sealant bead along the frame foot after the angle is installed.

### 7. Inspection requirements

This repair is interim and is limited to 6 000 flight cycles or twenty-four months from embodiment, whichever occurs first, after which the fitting must be replaced. Do a repeat dye penetrant inspection of the stop drill hole and of the angle perimeter every 1 500 flight cycles. Any growth from the stop drill hole makes the repair invalid immediately.

8. Record the repair against the aircraft and notify Aeralis within ten days.

## 9. Repair data

**Table 1 Repair data**

| | |
|---|---|
| Repair number | 53-21-13-003 |
| *Category* | Interim |
| *Damage type* | *Crack* |
| Fastener pattern | Two rows of 18 fasteners at 18 mm pitch and 12 mm edge distance measured from the sealant land, staggered, with the exclusion boundary on sheet 2 left unfastened |
| NDT method | Dye penetrant |
| Static strength restored | 85 % |
| Mass penalty | 0.9 kg |
| Aerodynamic smoothness | Not applicable, internal repair |
| Aeralis approval required | Yes |
| Repair drawings | RD-53-21-13-003 (2 sheets) |

## 4. Close-up

Interim repair limited to 6 000 flight cycles or twenty-four months; repeat dye penetrant inspection every 1 500 flight cycles.

Restore the external protection scheme and apply the repair stencil.